

Jiaxiang Liu

ARTIFICIAL INTELLIGENCE RESEARCHER

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Summary

Jiaxiang Liu is a Postdoctoral Researcher at GDIIST, advised by Prof. Luping Shi (Tsinghua). He received his Ph.D. from Zhejiang University in 2025 under Prof. Zuozhu Liu, and was previously a Research Intern at A*STAR Singapore with Prof. Joey Tianyi Zhou and Dr. Jiawei Du. His research focuses on **trustworthy multimodal AI for medicine**, building reliable medical vision-language models via *multi-expert chain-of-thought reasoning* and *adversarial robustness*, extending to brain decoding.

He has authored **34 peer-reviewed papers** — **10 independent first-author**, **8 co-first**, **3 corresponding**, and 13 co-author — including Cell Press *Patterns* (1, Cell Press journal, IF 10.8), *Nature Communications* (1, Nature Portfolio journal, IF 18.1), ICML (1), KDD (1), NeurIPS (1), ECCV (1), AAAI (3), EMNLP (2), MICCAI (2), ACL (1) and IEEE TIP / TCSVT (2, IF 11.1) / TMM / TIM / TETCI (6). His **Cell Press *Patterns*** paper presented the first clinically applicable pipeline for 3D fusion and *crown-root-bone* reconstruction, featured by Cell Press; two issued patents (IP Australia + Chinese). **H-index 16**, **i10-index 22** (Google Scholar).

Honours include the **National Postgraduate Scholarship**, First Prize for Outstanding Practical Achievements at Zhejiang University, Excellent Practical Achievements of Zhejiang Province, Jiangsu Jicui Postgraduate Scholarship (4 yrs), and Ni Jihong Prize; participated in **multiple NSFC and National Key R&D grants**. Reviewer for ICML (silver, 2026), NeurIPS, ICLR, KDD, CVPR, ECCV, MICCAI, AAAI, IEEE TCSVT/TIM.

Education

Zhejiang University

PH.D. IN ARTIFICIAL INTELLIGENCE, COLLEGE OF COMPUTER SCIENCE AND TECHNOLOGY

Hangzhou, China

2021 - 2025

- Supervisor: Prof. Zuozhu Liu.

Jiangxi University of Finance and Economics

M.S. IN COMPUTER TECHNOLOGY

Nanchang, China

2018 - 2021

- Supervisor: Prof. Yong Yang.

Jiangxi Normal University

B.S. IN INTERNET OF THINGS, SCHOOL OF COMPUTER INFORMATION ENGINEERING

Nanchang, China

2014 - 2018

Work Experience

Guangdong Institute of Intelligence Science and Technology (GDIIST)

POSTDOCTORAL RESEARCHER

Hengqin, Zhuhai, China

2025 - PRESENT

- Supervisor: Prof. Luping Shi (Tsinghua University).

Agency for Science, Technology and Research (A*STAR)

RESEARCH INTERN

Singapore

Sep. 2023 - Sep. 2024

- Supervisors: Prof. Joey Tianyi Zhou and Dr. Jiawei Du.

Publications

First Author (Independent) × 10

(1 Cell Press journal, 6 CCF-A / TH-CPL-A conferences, 2 JCR Q1 journals, 1 flagship medical conference)

Co-First Author × 8 | **Corresponding Author × 3** | **Co-Author × 13** (incl. 1 Nature Communications)

Symbols: # co-first author, * corresponding author.

SELECTED FIRST-AUTHOR CONFERENCE PAPERS

BioMedVR: Confusion-Aware Mixture-of-Prompt Experts for Biomedical Visual Reprogramming [PDF]

ECCV 2026

JIA XIANG LIU, TIAN XIANG HU, JU WEI GUAN, YU JIE WU, YU SONG WANG, YAO MU, ZUO ZHU LIU, MING KUN XU

2026

CCF-B / TH-CPL-A. First input-space dual-expert visual reprogramming for biomedical VLMs.

Self-Calibrated Consistency can Fight Back for Adversarial Robustness in Vision-Language Models [PDF] JIAXIANG LIU , JIAWEI DU, XIAO LIU, SHANGYANG LI, SONGCHEN MA, CHANGSHUO WANG, PRAYAG TIWARI, MINGKUN XU CCF-A / TH-CPL-A. arXiv:2510.22785; code at github.com/JXLiu-AI/SCC	ICML 2026 2026
MindAdapter: Few-Shot Parameter-Efficient Residual Calibration of Cross-Subject Brain-to-Visual Decoding Models [PDF] JIAXIANG LIU , JIAWEI DU, XUPENG CHEN, GUOQI LI, JIANG CAI, SIMON FONG, MINGKUN XU CCF-A / TH-CPL-A. Project page: jxliu-ai.github.io/MindAdapter	KDD 2026 (AI4Sciences track) 2026
KPL: Training-Free Medical Knowledge Mining of Vision-Language Models [PDF] JIAXIANG LIU , TIANXIANG HU, JIAWEI DU, RUIYUAN ZHANG, JOEY TIANYI ZHOU, ZUOZHU LIU CCF-A. Code: github.com/JXLiu-AI/KPL	AAAI 2025 (Oral) 2025
VPL: Visual Proxy Learning Framework for Zero-Shot Medical Image Diagnosis [PDF] JIAXIANG LIU , TIANXIANG HU, HUIMIN XIONG, JIAWEI DU, YANG FENG, JIAN WU, JOEY TIANYI ZHOU, ZUOZHU LIU TH-CPL / CAAI A	EMNLP 2024 2024
MedCoT: Medical Chain of Thought via Hierarchical Expert [PDF] JIAXIANG LIU , YUAN WANG, JIAWEI DU, JOEY TIANYI ZHOU, ZUOZHU LIU TH-CPL / CAAI A. Code: github.com/JXLiu-AI/MedCoT	EMNLP 2024 (Oral) 2024
ToothSegNet: Image Degradation Meets Tooth Segmentation in CBCT Images [PDF] JIAXIANG LIU , TIANXIANG HU, YANG FENG, WANGHUI DING, ZUOZHU LIU Flagship medical conference. Code: github.com/JXLiu-AI/Tooth-UNet	IEEE ISBI 2022
FIRST-AUTHOR JOURNAL PAPERS	
Deep learning-enabled 3D multimodal fusion of cone-beam CT and intraoral mesh scans for clinically applicable tooth-bone reconstruction [PDF] JIAXIANG LIU , JIN HAO, HANGZHENG LIN, WEI PAN, JIANFEI YANG, YANG FENG, GAOANG WANG, JIN LI, ZHIHE ZHAO, ZUOZHU LIU Cell Press data-science journal	Cell Press Patterns 2023
Parameter-Efficient Transfer Learning for Medical Visual Question Answering [PDF] JIAXIANG LIU , TIANXIANG HU, YAN ZHANG, YANG FENG, JIN HAO, JUNHUI LV, ZUOZHU LIU JCR Q1.	IEEE TETCI 2023
IFCoT: Insights Fusion Chain of Thought for Multimodal Reasoning in LLMs JIAXIANG LIU , JIAWEI DU, TIANXIANG HU, LINJIE TONG, YAN ZHANG, BING LI, YANG FENG, JIAN WU, JOEY TIANYI ZHOU, ZUOZHU LIU JCR Q1 / SCI Q1. IF 11.1	IEEE TCSVT (Accepted) 2026
Modest-Align: Data-Efficient Alignment for Vision-Language Models [PDF] JIAXIANG LIU , YUAN WANG, JIAWEI DU, JOEY TIANYI ZHOU, MINGKUN XU, ZUOZHU LIU JCR Q1. arXiv:2510.21606; code: github.com/JXLiu-AI/Modest-Align	Pattern Recognition (Under Review) 2025
CO-FIRST AUTHOR / CORRESPONDING AUTHOR PAPERS (SELECTED)	
3D-RAD: A Comprehensive 3D Radiology Med-VQA Dataset with Multi-Temporal Analysis and Diverse Diagnostic Tasks [PDF] XIAOTANG GAI, JIAXIANG LIU #, YU LI, ZIJIE MENG, JIAN WU, ZUOZHU LIU CCF-A / TH-CPL-A. Co-first author	NeurIPS 2026 (Dataset Track) 2026
Med-GLIP: Advancing Medical Language-Image Pre-training with Large-scale Grounded Dataset [PDF] ZIYE DENG, RUIHAN HE, JIAXIANG LIU #, YUAN WANG, ZIJIE MENG, SONGTAO JIANG, YONG XIE, ZUOZHU LIU CCF-B. Co-first author	IEEE BIBM 2025
GTCL: Graph-Text Contrastive Learning Meets Log Anomaly Detection [PDF] XIAO TAN, JIAXIANG LIU #, YUJIE WU, MINGKUN XU CCF-B. Co-first author	IEEE ICASSP 2026
Fair-MoE: Fairness-Oriented Mixture of Experts in Vision-Language [PDF] LINJIE TONG, PEIRAN WANG, JIAXIANG LIU *, ZUOZHU LIU CCF-B. Corresponding author	MICCAI 2025

ECLB: Efficient Contrastive Learning on Bi-Level for Noisy Labels

JUWEI GUAN, JIAXIANG LIU, SHUYING HUANG, YONG YANG
JCR Q1 / SCI Q1.

Knowledge-Based Systems (KBS)
2024

A ChatGPT-Aided Explainable Framework for Zero-Shot Medical Image Diagnosis [PDF]

JIAXIANG LIU, TIANXIANG HU, YAN ZHANG, XIAOTANG GAI, YANG FENG, ZUOZHU LIU

ICML-IMLH Workshop
2023

Patents

Multiple Decision Maps Based Infrared and Visible Image Fusion

YONG YANG (SUPERVISOR); LIU, JIAXIANG; ET AL.

IP Australia, No. 2020100178
2021

Image Fusion Method, Device, Equipment and Readable Storage Medium

YANG FENG; ZUOZHU LIU; LIU, JIAXIANG

Chinese Patent, No. 202311022450
2023

Research Projects

Multi-Expert Verified Medical Chain-of-Thought Reasoning (MedCoT framework)

SCIENTIFIC RESEARCH FUND OF ZHEJIANG UNIVERSITY

Hangzhou, China
2023 - 2025

- Hierarchical experts + Sparse MoE + feature-level voting; a 256M model outperforms LLaVA-Med (7B) by 5%+.
- **Mentoring**: mentored 2 master students, who published 4 papers at **NeurIPS, NAACL, AAAI, MICCAI**.
- **Personal outputs**: 4 top-tier conference papers as lead/co-lead author (**MedCoT, KPL, VPL, MedThink**); awarded **First Prize** at Zhejiang University and the **Outstanding Practical Achievement Award** of Zhejiang Province

Clinical applicable multimodal fusion algorithm research for 3D intraoral scanning and CBCT data

SCIENTIFIC RESEARCH FUND OF ZHEJIANG UNIVERSITY

2022 - 2023

- Reduced multimodal oral data *crown-root-bone* reconstruction time from a conventional **5-7 hours** to **20-25 minutes** (~15x speed-up).
- Algorithm deployed in the **Angelalign IRS** clinical orthodontic system; results featured by **Cell Press** in an official news release.
- **Mentoring**: mentored 1 undergraduate student, who published 3 papers at **MIDL, MICCAI, BIBM**.
- **Personal outputs**: **Cell Press Patterns 2023** first-author paper; awarded **First Prize** at Zhejiang University and **Excellent Practical Achievement** of Zhejiang Province

Infrared and visible image fusion for public safety

STUDENT RESEARCH PROJECT OF JIANGXI UNIVERSITY OF FINANCE AND ECONOMICS

Nanchang, China
2020 - 2021

- Designed multi-scale and modal feature fusion networks for IR-VIS image fusion in public-safety scenarios.
- Outputs: 4 IEEE journal / conference papers (TCSVT, TMM, TIM, ICME Oral) and 1 international patent (IP Australia).

Skills & Service

Programming	Python, MATLAB, C, HTML5, SQL.
Languages	Native Chinese; Fluent English.
Editorial service	Young Editorial Board Member, <i>Computer and Modernization</i> .
Conference reviewer	ICML (silver tier, 2026), NeurIPS, ICLR, KDD, CVPR, ECCV, MICCAI, AAAI.
Journal reviewer	IEEE TCSVT, IEEE TIM, IEEE TIP.

Honors & Awards

2024	First Prize (university level) , Outstanding Practical Achievements of Professional Degree Graduate Students, Zhejiang University	Hangzhou, China
2024	Provincial Award , Outstanding Practical Achievements of Professional Degree Graduate Students, Zhejiang Province	Zhejiang, China
2022-2025	Multiple Graduate Scholarships , Jiangsu Province Jicui Postgraduate Scholarship (4 yrs) & Ni Jihong Prize, Zhejiang University	China
2021	National Scholarship , National Scholarship for Postgraduates	China
2019-2020	National & Provincial 1st-3rd Prizes (5 total) , Huawei Cup China Post-Graduate Mathematical Modeling (16th 2nd; 17th 3rd) & Jiangxi Province Modeling (1st, 2nd) & 7th TipDM Cup Data Mining (2nd)	China
2018-2025	Excellent Graduate , Jiangxi Normal University & Zhejiang University	China
2015	Provincial First Prize , Jiangxi Province University Student Science and Technology Innovation and Skills Competition	Jiangxi, China